

# Thomas Gozalie

Backend / AI Engineer

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## About Me

I am an aspiring backend and AI engineer who applies research rigor and best practices from deep learning to build simple, extensible backend systems. As a Junior Researcher, I have hands-on experience with designing and implementing Computer Vision experiments with PyTorch that pushed the state-of-the-art on two geosensing computer vision datasets while documenting results in technical papers and communicating effectively with my co-authors. My experience on Backend Development with Node.js and NestJS enables me to develop API endpoints for web and mobile-facing applications with clean organization and architecture. I aim to bring this research mindset and my commitment to lifelong learning to a Backend Engineer and/or AI Engineer role that values practical, scalable solutions.

## Education

**Bina Nusantara University (BINUS) | Bachelor of Computer Science**

Sep, 2022 - Jun, 2026

GPA: 3.84 (Semester 7)

**Programming Languages:** Python, Java, C, C++, SQL

**Notable classes:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Competitive Programming, Database Technology, Software Engineering, Computer Networks

**RevoU | Full-Stack Software Engineering**

Jun, 2025 - Jan, 2026

Certificate of Excellence

**Skills:** TypeScript, Next.js, TailwindCSS, React, React Native, Jest, Postman, Git, GitHub

**Final Project:** PlatePal

## Working Experiences

**BINUS AI Research and Development Center | Junior Researcher**

Sep, 2025 - Present

- Adapted Transformer-Mamba hybrid architecture (MambaVision) for change detection, outperforming the state-of-the-art on LEVIR-CD and WHU-CD with up to +0.92 IoU increase and 68.5% faster throughput by adding LayerNorm2d and hierarchical feature extraction to MambaVision with PyTorch to improve multi-scale spatial analysis. Published at ICCRAIDS 2026, at <https://ieeexplore.ieee.org/document/11519636>.
- Developed a modular dataset pipeline for LEVIR-CD and WHU-CD with PyTorch and scikit-learn, incorporating custom patchification and augmentation techniques to handle very high-resolution (VHR) remote sensing data and improve model's test set performance.
- Implemented deep learning model architectures encompassing Siamese U-Net, Vision Transformers and Vision Mamba, data preprocessing pipelines and experiments for geospatial land-cover change detection using PyTorch and PyTorch Lightning.
- Utilized Docker and Ansible for setting up reproducible environments for experiments that can run on any platform with immutable deployment principle, wrote Dockerfile for scripting experiment container environment setup from a pre-built PyTorch image.

## Project Experiences

**PlatePal**

Dec, 2025 - Jan, 2026

PlatePal is an AI-powered nutrition tracking mobile application that can automatically extract nutrition information from a meal picture by leveraging LLMs, built with React Native + Expo user-facing frontend, Next.js and shadcn/ui admin panel, with a hosted NestJS backend utilizing PostgreSQL database and Prisma ORM. Aside from the MVP, I have implemented secure authentication and role-based access control using JWT strategies, and managed access and refresh token for user session security while minimizing authorization friction. Finally, I documented a collection of REST API requests for PlatePal with Postman and Postman Documenter.

## Competitions

**Data Slayer 3.0**

Nov, 2025

- Placed #2 among solo contestants, #19 out of 163 teams in the official ranking, and wait-listed for finals.
- Obtained 84% accuracy by classifying awake/drowsy driver video footage on the unseen test set by designing a machine learning pipeline for driver face localization with YOLO-face, face expression encoding by training an ImageNet-pretrained image encoder implemented with PyTorch and OpenCV, and sequence classification into distinct awake/drowsy labels with a temporal convolution model.
- Processed a noisy label dataset through data selection, manual labeling and image augmentation to guide the model into capturing face expression and temporal features.

**INC 2024**

Nov, 2024

- Solved 4 Competitive Programming problems during the competition, achieving Top 10% percentile of teams from the Indonesian region.
- Utilized mathematical, greedy and dynamic programming methods in C++ for solving problems, and analyzed Big-O complexity to determine time and space complexity upper-bound for submitted solutions.

## Additional Information

**Hard Skills:**

- Programming Languages:** Python, JavaScript, Java, C, C++
- Deep Learning / AI:** Python, PyTorch, OpenCV, scikit-learn
- Data Visualization:** Matplotlib, Pandas
- Backend Development:** TypeScript, NestJS, Prisma, PostgreSQL, Postman
- Frontend Development:** Next.js, React.js, React Native, TailwindCSS, shadcn/ui
- CI / CD:** Git, GitHub, Docker, Ansible

**Soft Skills:** Scientific Writing, Research, Collaboration

**Languages:** Indonesian, English

**Certifications:** 1. [RevoU Full-Stack Software Engineering Certificate of Excellence](#)

06 Feb, 2026